

Chem 273 Discussion 4

Logistics

- Lecture Exercise 4 is due next Monday, the 15th
- Problem Set 1 is due next Thursday, the 18th
 - As a reminder, Problem Sets are meant to be a bit longer than Lecture Exercises
- Lecture Exercise 2 grades were released.

Function Review

```
def Tiles(N = 100):
```

- What does $N = 100$ mean?
- How would you call this function with $N = 500$, $N = 2000$, $N = 5$, etc.?

Function Review

Define a function ONCE, then call it as many times as you would like

```
def Tiles(N = 100):
```



Don't change this value!

```
Tiles(100)
```

```
Tiles(500)
```

```
Tiles(4)
```

```
Tiles(10000000000)
```

Indexing

```
V = np.random.uniform(0,1,(N,1))  
D = np.zeros((N,N))
```

What data type are V or D?

2D NumPy Array

```
X = np.array([[2, 3, 4],[6, 1, 5]])  
print(X)
```

```
[[2 3 4]  
 [6 1 5]]
```

How would you index into the first row and first column?

`X[0, 0]`

`X[0][0]`

Indexing

“Chained Indexing” is something like this: `X[0][0]`

```
[[2 3 4]
 [6 1 5]]
```

This works, but what happens is two steps:

`X[0]` = will return a 1D array

```
array([2, 3, 4])
```

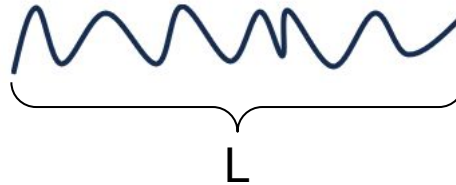
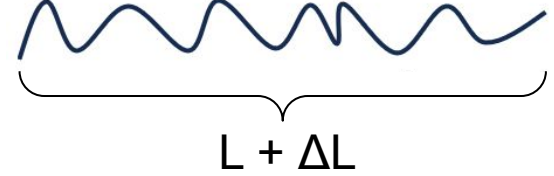
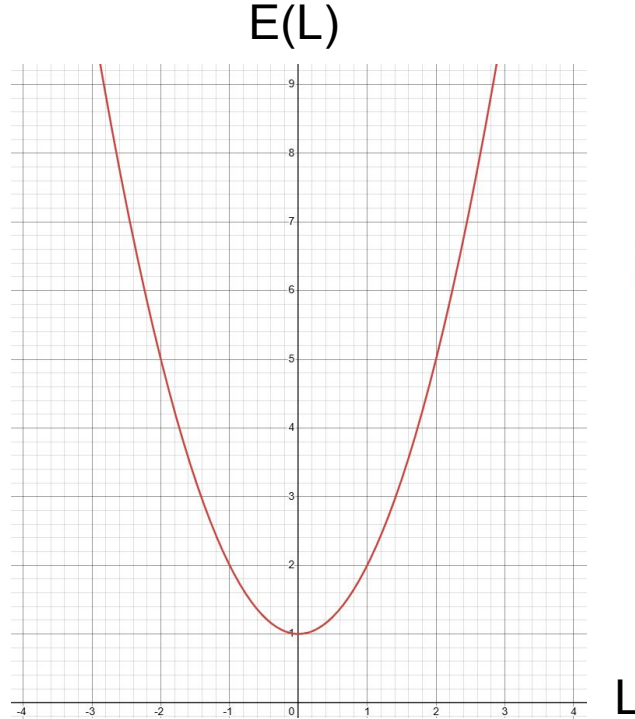
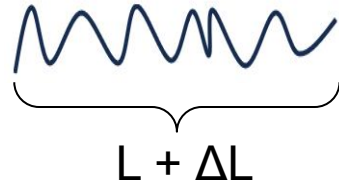
`[0]` = indexing into that row

```
2
```

Instead to easily get a value at a given set of indices, use `X[0, 0]` which is more efficient and faster in NumPy

`X[0][0] == X[0, 0]` but `X[0, 0]` is faster

Lecture 3 Assignment Solution



Applying the Taylor series

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

$$f(x) - f(x_0) = f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

$$f(x) = E(L + \Delta L) \quad f(x_0) = E(L)$$

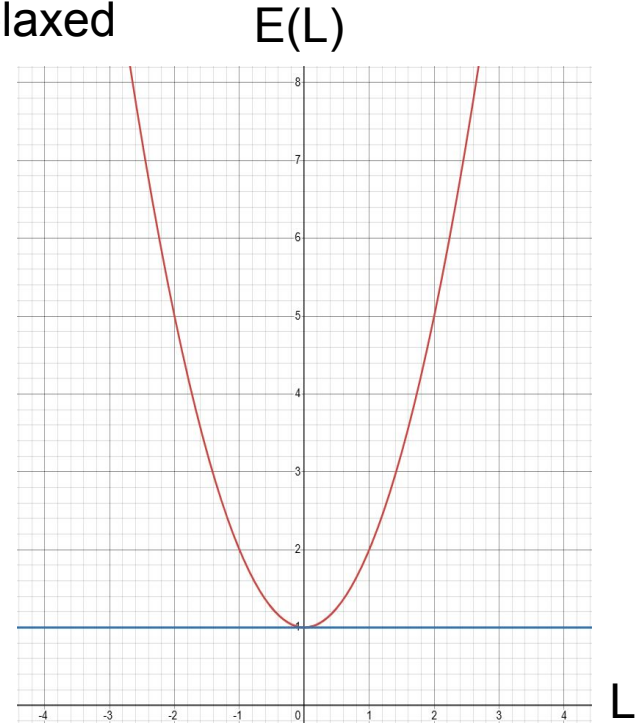
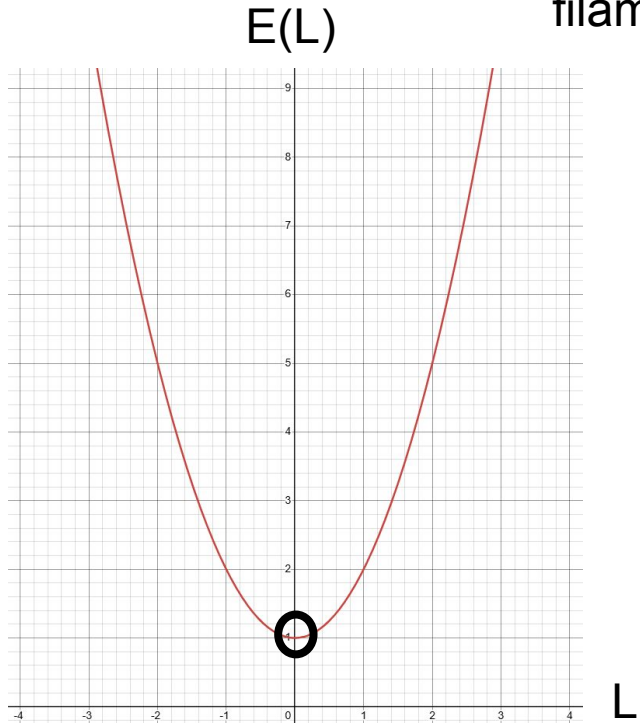
Applying the Taylor series

$$E(L + \Delta L) - E(L) = f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

$$\Delta E = f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

Applying the Taylor series

At EQUILIBRIUM the actin filament is completely relaxed



Applying the Taylor series

$$E(L + \Delta L) - E(L) = f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

$$\Delta E = f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

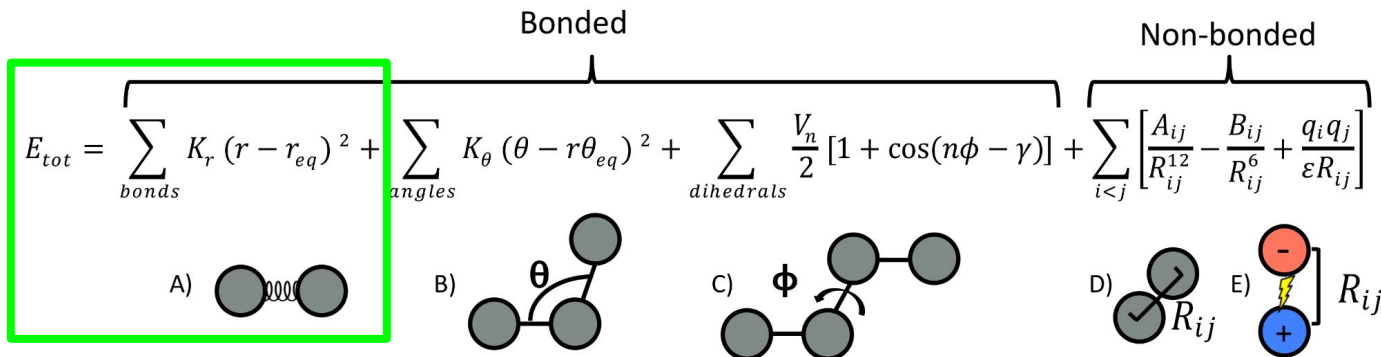
$$\Delta E = 0 + \frac{f''(x_0)}{2}(x - x_0)^2$$

Applying the Taylor series

We just derived hooke's law for
spring energy!!!

$$\Delta E = \frac{1}{2} f''(x_0)(x - x_0)^2$$

$$\Delta E = \frac{1}{2} k (x - x_0)^2$$



Integral Practice

$$\int 12 \, dy$$

$$12y + c$$

Integral Practice

$$\int 4x^6 - 2x^3 + 7x - 4 \, dx$$

$$\frac{4}{7}x^7 - \frac{1}{2}x^4 + \frac{7}{2}x^2 - 4x + c$$

Integral Practice

$$\int 2 \sin(x) - \cos(x) dx$$

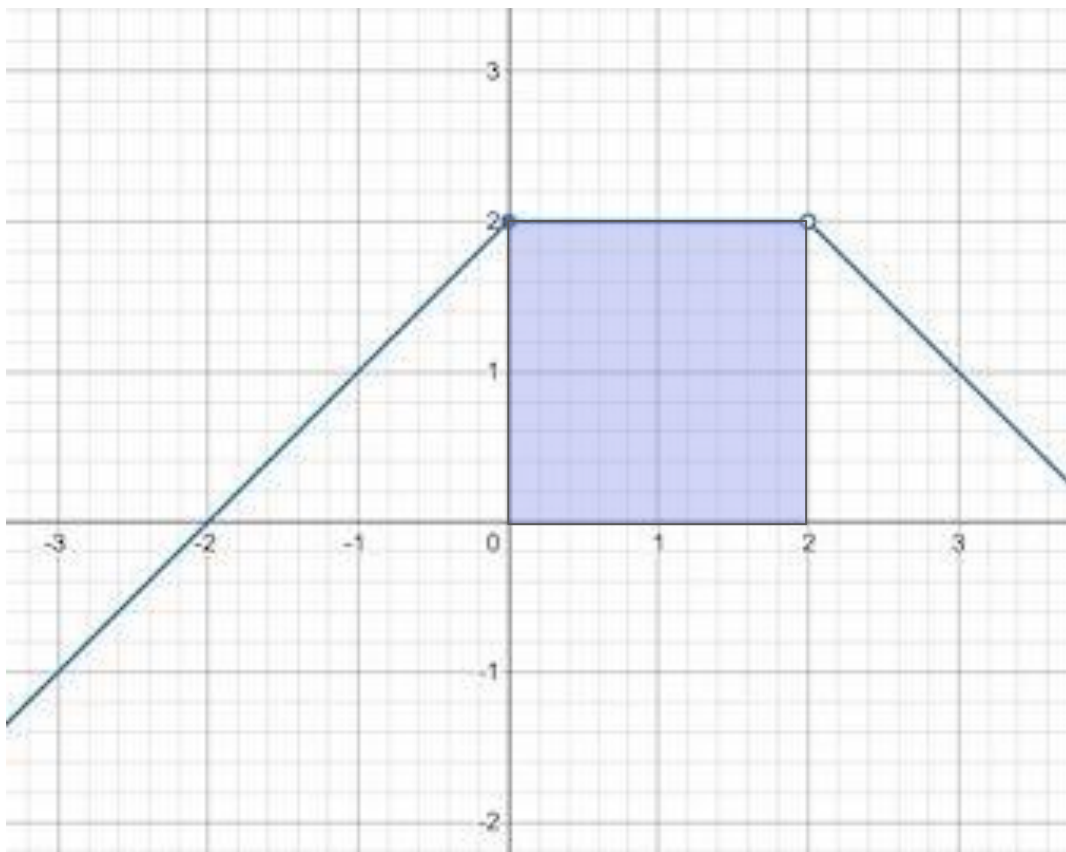
$$2 \int \sin(x) dx - \int \cos(x) dx$$

$$- 2 \cos(x) - \sin(x) + c$$

Integral Practice

$$\int_0^2 f(x) dx$$

4



Integral Practice

$$\int_1^6 12x^3 - 9x^2 + 2 \, dx$$

$$(3x^4 - 3x^3 + 2x) \Big|_1^6$$

$$(3(6)^4 - 3(6)^3 + 2(6)) - (3(1)^4 - 3(1)^3 + 2(1))$$

$$3252 - 2 = \boxed{3250}$$

Integral Practice

$$\int 4x + 25y^2 dx$$

$$2x^2 + 25y^2x + c$$

Integral Practice

Determine $f(x)$ given that $f'(x) = 12x^2 - 4x$ and $f(-3) = 17$.

$$f(x) = \int 12x^2 - 4x \, dx$$

$$4x^3 - 2x^2 + c$$

$$17 = f(-3) = 4(-3)^3 - 2(-3)^2 + c \qquad c = 143$$

$$f(x) = 4x^3 - 2x^2 + 143$$